AI-Ready Data: Knowledge Extraction from Archival Lab Notebooks

IEEE Big Data 2024, CAS Workshop

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December 17, 2024

DREXEL UNIVERSITY Metadata Research Center College of Computing & Informatics

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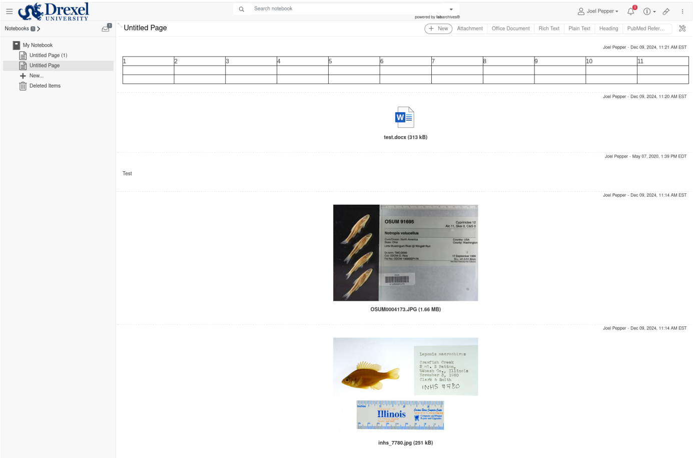
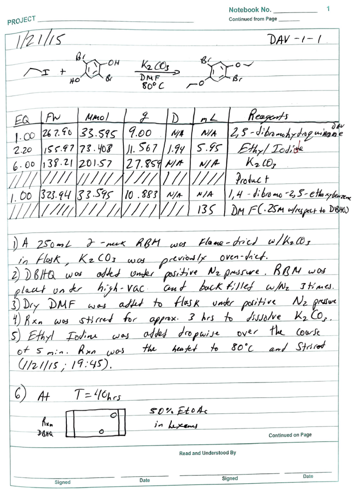
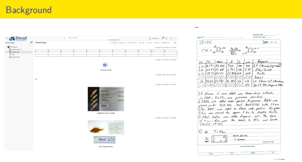
Background



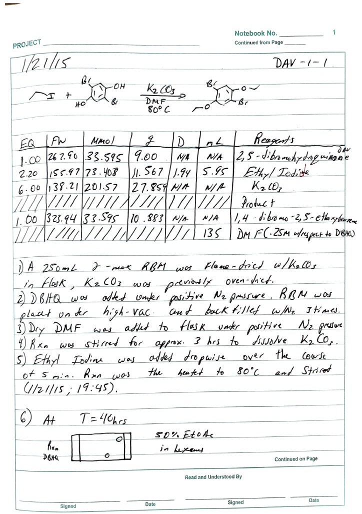




1. Over the past couple decades, traditional paper-based laboratory notebooks have been increasingly replaced by electronic lab notebooks, aka ELNs. ELNs are able to streamline workflows by allowing real-time data entry, integration with lab instruments, and direct linking to files and databases.
2. ELNs can help maintain archival integrity, by helping standardize records, minimizing data loss, and enabling easier sharing of methods and results between scientists.
3. Talk about pictures of notebooks, step slide

Background

- Gloves, chemicals, nature of work, complex reactions and custom diagrams make switch to digital notebooks unfeasible for chemists
- Notes recorded on special, chemical resistant paper
- Manual logging of paper notes as faithful digital copies extremely time intensive



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AI-Ready Archival Lab Notebooks
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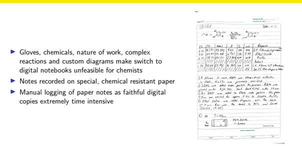
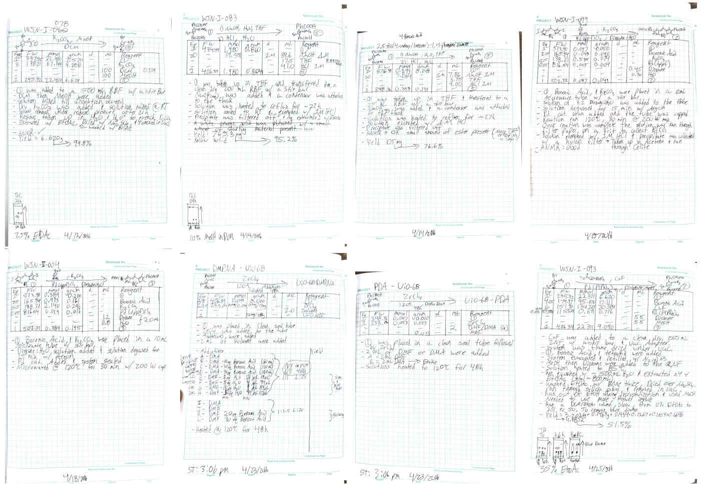


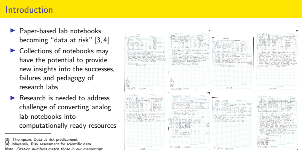
1. While the transition from paper to electronic notebooks has been fairly painless for some experimentalists, chemists in particular face a number of challenges in making the switch. Gloves, chemicals, the very hands on and in some ways potentially “messy” nature of their work, the complex reactions that often need written out, and the custom diagrams that are used to define experiments can make the switch to digital notebooks unfeasible.
2. Notes are often recorded by chemists on special, chemical resistant paper to account for spills, which would of course destroy any bench-side laptop.
3. And additionally, the manual logging of paper notes as faithful digital copies in ELN software is extremely time intensive, and not an undertaking most labs will consider worth pursuing.

Introduction

- Paper-based lab notebooks becoming "data at risk" [3, 4]
- Collections of notebooks may have the potential to provide new insights into the successes, failures and pedagogy of research labs
- Research is needed to address challenge of converting analog lab notebooks into computationally ready resources



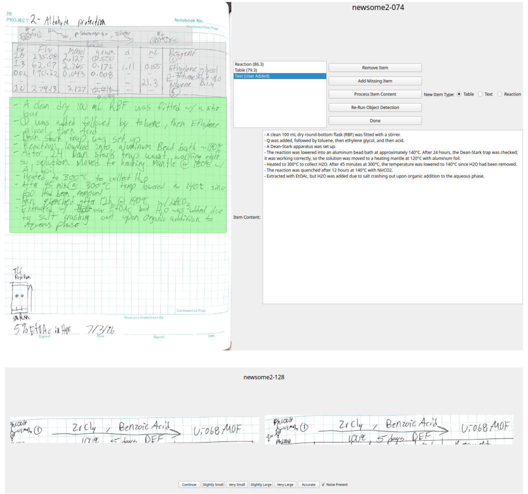
[3]: Thompson, Data-at-risk predicament
[4]: Mayernik, Risk assessment for scientific data
Note: Citation numbers match those in our manuscript



1. Being analog records in a decidedly digital world, paper-based lab notebooks are becoming "data at risk"
2. Why this matters is that collections of notebooks, from one discrete research group or in larger aggregate, have the potential to provide new insights into the successes, failures and pedagogy of research labs
3. Research is needed to address the challenge of converting analog lab notebooks into computationally ready resources, in order to enable large scale computational analysis of their contents

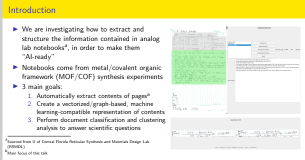
Introduction

- We are investigating how to extract and structure the information contained in analog lab notebooks[a], in order to make them "AI-ready"
- Notebooks come from metal/covalent organic framework (MOF/COF) synthesis experiments
- 3 main goals:
  1. Automatically extract contents of pages[b]
  2. Create a vectorized/graph-based, machine learning-compatible representation of contents
  3. Perform document classification and clustering analysis to answer scientific questions



[a] Sourced from U of Central Florida Reticular Synthesis and Materials Design Lab (RSMDL)

[b] Main focus of this talk

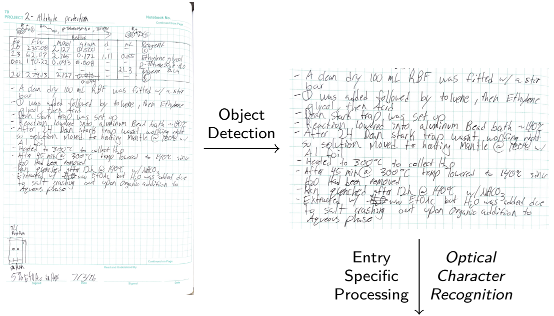


1. We are investigating how to extract and structure the information contained within analog lab notebooks, in order to make them AI-ready. The notebooks in question come from the University of Central Florida Reticular Synthesis and Materials Design Lab, with whom we at the Drexel Metadata Research Center are collaborating on this project.
2. The chemistry lab notebooks come from metal/covalent organic framework (MOF/COF) synthesis experiments
3. The work we are presenting is preliminary and focuses on only the first of these, but long term our project has 3 main goals:
4. To automatically extract the contents of pages
5. Then subsequently, to create a vectorized/graph-based, machine learning-compatible representation of their contents
6. And finally, to perform document classification and clustering analysis to answer scientific questions, such as what yet unseen patterns might differentiate successful experiments from failures

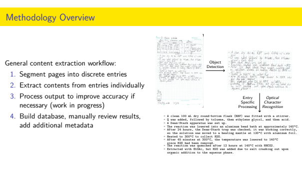
Methodology Overview

General content extraction workflow:

1. Segment pages into discrete entries
2. Extract contents from entries individually
3. Process output to improve accuracy if necessary (work in progress)
4. Build database, manually review results, add additional metadata



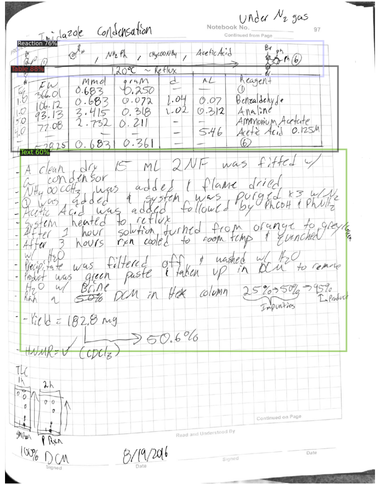
- A clean 100 mL dry round-bottom flask (RBF) was fitted with a stirrer.
- Q was added, followed by toluene, then ethylene glycol, and then acid.
- A Dean-Stark apparatus was set up.
- The reaction was lowered into an aluminum bead bath at approximately 140°C.
- After 24 hours, the Dean-Stark trap was checked; it was working correctly, so the solution was moved to a heating mantle at 120°C with aluminum foil.
- Heated to 300°C to collect H2O.
- After 45 minutes at 300°C, the temperature was lowered to 140°C since H2O had been removed.
- The reaction was quenched after 12 hours at 140°C with NHCO2.
- Extracted with EtOAc, but H2O was added due to salt crashing out upon organic addition to the aqueous phase.



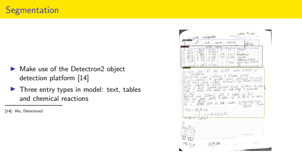
1. The general workflow for the content extraction portion of this is: to first segment pages into different entry types–currently tables, text blocks and reactions
2. Extract the contents from entries individually, using entry type specific processes
3. Further post-process the outputs to improve accuracy, if necessary
4. Build a database collating the outputs from all pages, manually review the results using a couple GUIs I put together, and manually add additional metadata to the database such as which page or pages correspond to what kinds of experiment

Segmentation

- Make use of the Detectron2 object detection platform [14]
- Three entry types in model: text, tables and chemical reactions



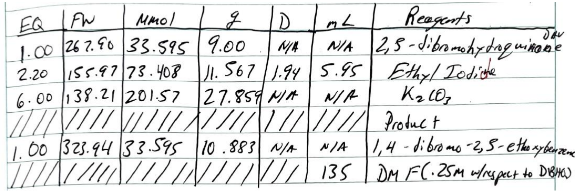
[14]: Wu, Detectron2



1. In order to perform automated page segmentation, we make use of the Detectron2 object detection platform
2. After reviewing the notebooks in our study, we determined there to be three major entry types worth focusing on: text, tables and chemical reactions. Some calculations and diagrams found in the notebooks were excluded, as these entry types would likely be difficult to leverage in machine learning analysis.
3. We hand labeled 101 instances of tables, 101 reactions, and 153 text blocks from two RSMDL notebooks by two different authors, and used this dataset to train a model

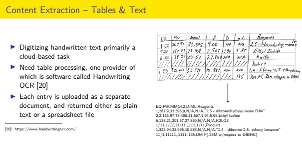
Content Extraction – Tables & Text

- Digitizing handwritten text primarily a cloud-based task
- Need table processing, one provider of which is software called Handwriting OCR [20]
- Each entry is uploaded as a separate document, and returned either as plain text or a spreadsheet file



EQ,FW,MMOl,2,D,ML,Reagents
1,267.9,33.595,9,N/A,N/A,"2,5 - dibromohydroquinone DAV"
2.2,155.97,73.408,11.567,1.94,5.95,Ethyl Iodine
6,138.21,201.57,27.859,N/A,N/A,K2LO3
1/11,////,11/11.,,111,1/11,Product
1,323.94,33.595,10.883,N/A,N/A,"1,4 - dibromo-2,5- ethoxy benzene"
11/1,11111,,1111,,135,DM F(.25M w/respect to DIBHG)

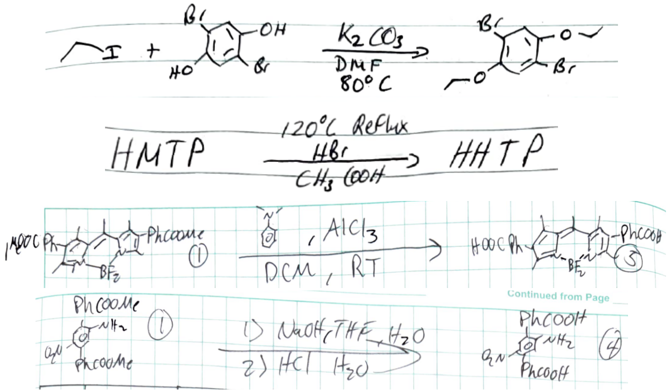
[20]: https://www.handwritingocr.com/

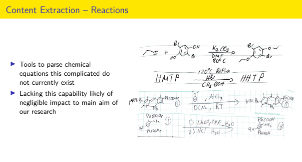


1. Unfortunately, open-source, locally executable tools for digitizing handwritten text lag fairly significantly behind commercial tools in accuracy and features
2. Cloud-based offerings are available from the usual suspects–Amazon, Google, Microsoft and OpenAI–as well as other smaller competitors
3. Of particular relevance to this project is the processing of tables, which requires borders to be interpreted as separators (i.e. commas) in the output.
4. A commercial, cloud based software which we are currently using is called Handwriting OCR, which is capable of handling both tabular inputs and plain text with an accuracy comparable to the other major cloud providers
5. Each entry is uploaded as a separate document, and returned either as plain text or a spreadsheet file
6. Talk about how to use ChatGPT it would need to separate the text and tables on the page, process tables separately, and ignore any text/characters in reactions and diagrams

Content Extraction – Reactions

- Tools to parse chemical equations this complicated do not currently exist
- Lacking this capability likely of negligible impact to main aim of our research

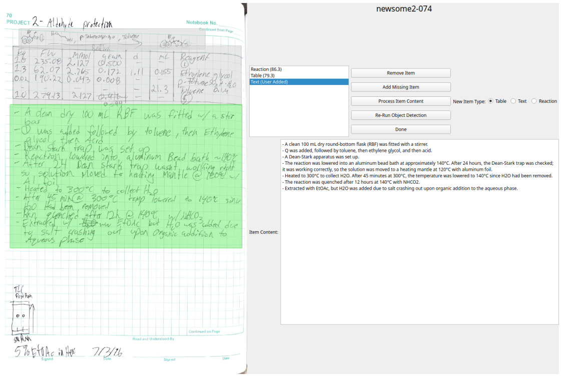


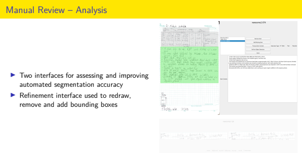


1. While there is certainly merit in the automated parsing of chemical equations, the tech needed for equations of this complexity does not currently exist.
2. Lacking this capability is likely of negligible impact to the main aim of our research, as most of the unique information we are looking to study is contained in the tables and text of the documents.
3. A small review of the tools that do exist for hand-drawn organic molecule parsing is included in our paper.

Manual Review – Analysis

- Two interfaces for assessing and improving automated segmentation accuracy
- Refinement interface used to redraw, remove and add bounding boxes

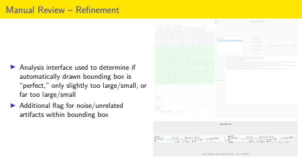




1. We have designed two interfaces to aid in assessing and improving automated segmentation accuracy. These interfaces also help assess the existing archival challenges that limit or confound the automated extraction of accurate information from the notebooks.
2. The first of these is the refinement interface, which allows a user to redraw bounding boxes with their mouse, remove spurious ones, and add missing ones. The extracted contents can also be manually refined. Both content versions are saved in a database for subsequent analysis and potential archiving.

Manual Review – Refinement

- Analysis interface used to determine if automatically drawn bounding box is "perfect," only slightly too large/small, or far too large/small
- Additional flag for noise/unrelated artifacts within bounding box



1. The second of these is the analysis interface which shows both the original and user-drawn versions of each entry where applicable. It allows a user to determine if the automatically drawn bounding box was only slightly too large/small and still likely sufficient for accurate content extraction, or far too large/small and very likely to cause processing errors. If an entry was added by the user, deleted by the user, or the user did not redraw the bounding box, then it is skipped in the interface and automatically recorded as deleted, added or accurate (i.e. perfect) in the database.
2. There is an additional flag to mark whether noise is present, noise referring to unrelated overlapping objects falling inevitably within the bounding box of an entry.

Results

- 154 pages for the testing set and manual review
- 78.8% of entries have accurate automated bounding boxes
- 15.6% of entries have nontrivial noise within their bounding boxes
- There are some experiment-specific diagrams that Detectron2 interpreted as tables
- Table style varied between the two authors
- Corrections in the notebooks very hard to automatically parse

Bounding Box Quality	Count
Perfect	41
Erroneous	53
Missed	50
Slightly Small	176
Slightly Large	81
Very Small	27
Very Large	3
Acceptable Quality	298
Unacceptable Quality	80
Erroneously Labeled	53

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1. We had 154 pages for the testing set and manual review
2. Excluding the erroneous segmentations, 298 out of 378 entries (78.8%) have automated bounding boxes that accurately capture their contents. 53 of the bounding boxes it produced were for nonexistent entries, and it missed 50 entries entirely. Notably, all of the 50 missed entries were text blocks, no reactions or tables were missed.
3. 59 entries have nontrivial noise within their bounding boxes, about 16%.
4. (Next page) There are experiment-specific diagrams in the notebooks, some of which are reminiscent of tables and confused Detectron2
5. (Next page) The authors drew tables using distinct styles. The primary difference that impacts analysis is that one author drew diagonal slashes in empty table records, and the other drew hyphens. The diagonal slashes are interpreted as 1's by the OCR software in some cases.
6. (Next page) And lastly, author correction and amendments are very difficult to parse automatically.

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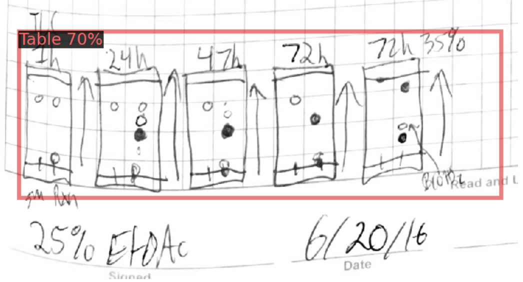

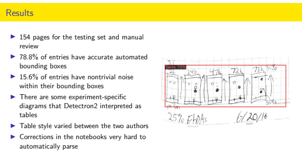

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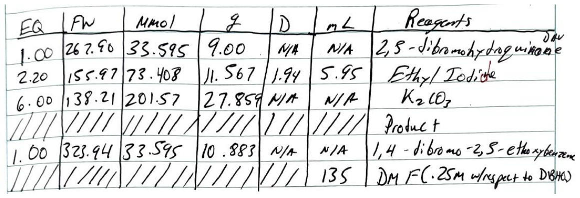

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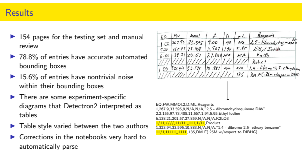
1,267.9,33.595,9,N/A,N/A,"2,5 - dibromohydroquinone DAV"
2.2,155.97,73.408,11.567,1.94,5.95,Ethyl Iodine
6,138.21,201.57,27.859,N/A,N/A,K2LO3
1/11,////,11/11.,,111,1/11,Product
1,323.94,33.595,10.883,N/A,N/A,"1,4 - dibromo-2,5- ethoxy benzene"
11/1,11111,,1111,,135,DM F(.25M w/respect to DIBHG)



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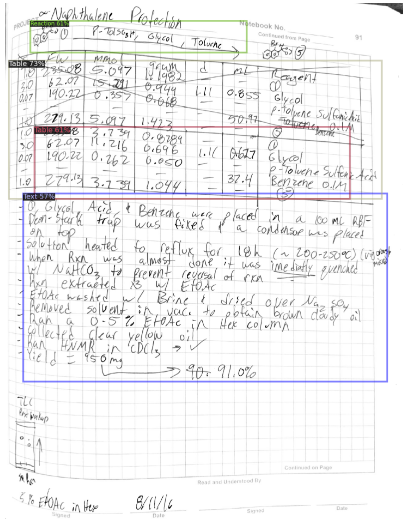

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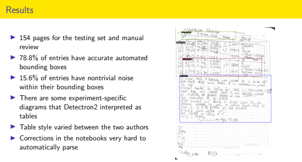

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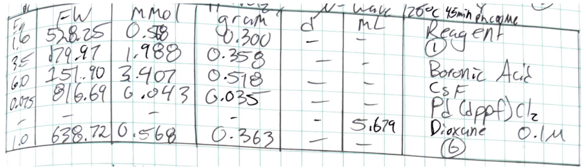
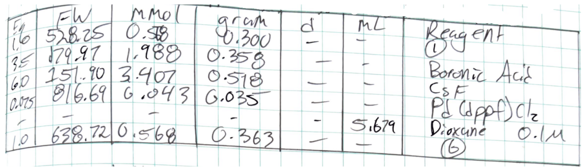
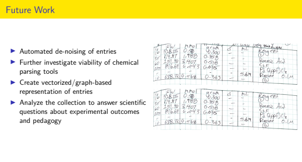
Future Work

- Automated de-noising of entries
- Further investigate viability of chemical parsing tools
- Create vectorized/graph-based representation of entries
- Analyze the collection to answer scientific questions about experimental outcomes and pedagogy



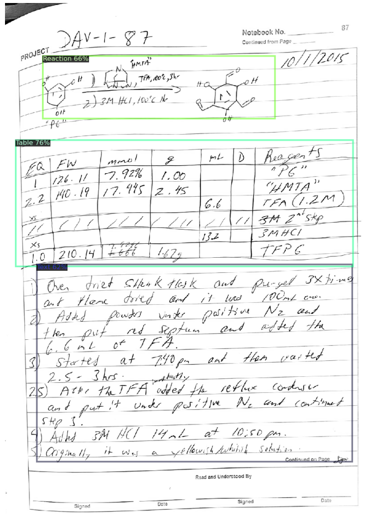


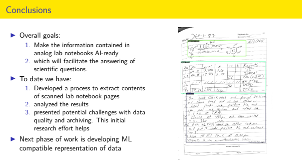


1. Quite a lot of future work still remains for this project, currently we are exploring the automated de-noising of entries using an auto encoder or similar (talk about figure)
2. We also plan to further investigate viability of chemical parsing tools to see if any could be adapted for full equations
3. The next major phase of this research is to create vectorized/graph-based representation of entries, which another PhD student is currently working on by using LLMs to tokenize the notebook text and build it into a graph representation
4. And ultimately, we aim to analyze the collection to answer scientific questions about experimental outcomes and pedagogy, such as:
5. Are there any yet undiscovered patterns in the experimental protocols that separate successful experiments from failed ones?
6. When is a student properly trained in these chemical syntheses, and can this information be automatically gleaned from their lab notebooks?

Conclusions

- Overall goals:
  1. Make the information contained in analog lab notebooks AI-ready
  2. which will facilitate the answering of scientific questions.
- To date we have:
  1. Developed a process to extract contents of scanned lab notebook pages
  2. analyzed the results
  3. presented potential challenges with data quality and archiving. This initial research effort helps
- Next phase of work is developing ML compatible representation of data





The overall goals of our work are: to advance approaches for making the information contained in analog lab notebooks AI-ready, and to use that data to answer of scientific questions. We have developed a process to extract the contents of scanned lab notebook pages, analyzed the results and presented potential challenges with data quality and archiving. The immediate next phase of our work is developing ML compatible representation of the extracted data.

Questions?

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Metadata Research Center
College of Computing & Informatics

   

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AI-Ready Archival Lab Notebooks
└Methodology
└Questions?

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